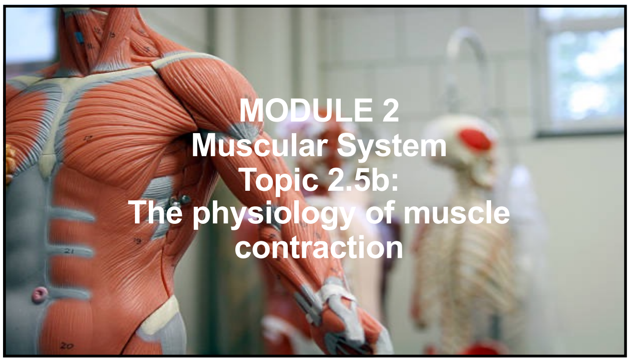

MODULE 2
Muscular System
Topic 2.5b:
The physiology of muscle contraction

Learning Objectives

- ✓ Define 'motor unit'
- ✓ Outline the difference between hypertrophy and hyperplasia (with respect to skeletal muscle enlargement)
- ✓ Briefly describe 3 biochemical pathways for skeletal muscle to generate ATP
- ✓ Name 3 types of muscle fibres and describe how they rely on different cellular processes to obtain ATP

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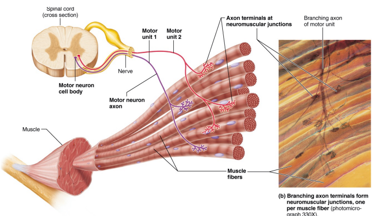
The Motor unit

- **Each muscle** = served one motor nerve
- **Motor nerve** contains axons
- **Motor unit** = 1 x motor neuron + muscle fibres (4-300 fibres/motor unit)
- **many fibres** = larger motor unit
  - *Strength (leg muscles)*
- **few fibres** = smaller motor unit
  - *fine control (fingers/eyes)*



a) Axons of motor neurons extend from the spinal cord to the muscle. At the muscle, each axon divides into a number of axon terminals that form neuromuscular junctions with muscle fibers scattered throughout the muscle.

(b) Branching axon terminals form neuromuscular junctions, one per muscle fiber (photomicrograph 330X).

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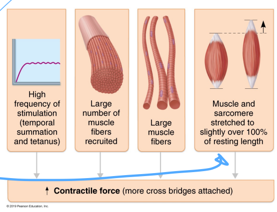
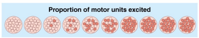
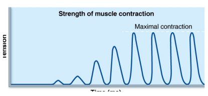
(size principle : smallest → biggest motor unit)

Muscle contraction
What affects it?

1. Frequency & strength of stimulus
2. Recruitment of motor units
3. Size of fibres
4. Degree of muscle stretch







Increased frequency of stimulus causes muscle contractions to become more forceful and build upon one another (summation) = "treppe effect" (German for 'staircase')

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Muscles
Exercise

Aerobic exercise (long distance running)

- improves O_2 delivery and utilisation by cells
- > capillaries, mitochondria and myoglobin = improves muscle metabolism

Resistance exercise e.g.: weight lifting (high intensity)

- **Hypertrophy** (hyper = ↑) = individual muscle fibre ***size* increases** (**NOT** an **increase in** muscle fibres) ↑ myofibrils & myofilaments
- **Hyperplasia** (hyper = ↑) = cell proliferation, splitting of fibres, ↑ myofilaments, connective tissue

Atrophy → *If you don't USE it, you LOSE it!* → muscle gets smaller

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Types of muscle contraction

Type A: *Isotonic* = change length (e.g. bicep curls)

i. **Concentric** = shortens (elbow flex)

ii. **Eccentric** = lengthens (elbow ext)

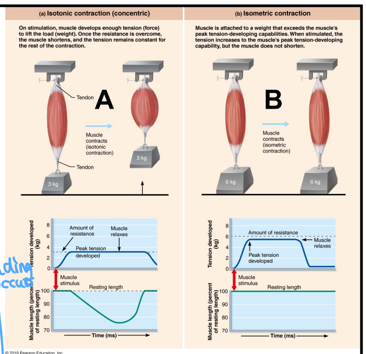
microtears & delayed soreness

Type B: ***Isometric*** ≠ change length

- Create tension BUT myofilaments **do NOT** sliding past each (no shortening)

binding occurs

load > muscle force (no movement)



Muscle Metabolism

limited ATP reserve

But first - What is metabolism?

- How muscle uses energy (ATP from glucose) to fuel contraction **[ATP = adenosine triphosphate]**
- There are **3** potential sources of ATP to power muscle contractions.

1. **Direct Phosphorylation** = Stored + Creatine Phosphate [CP] (15s)
2. **Anaerobically** = Stored ATP + CP + Glycolysis = (30-40s)
   - produces ATP 2.5x faster than aerobic pathway (↑ force)
3. **Aerobically =** Glucose + FFAs + AAs (> 60s) [FFA = free fatty acids; AA = amino acids]
   - 95% all ATP used for muscle activity from aerobic respiration (endurance, ↓ force)

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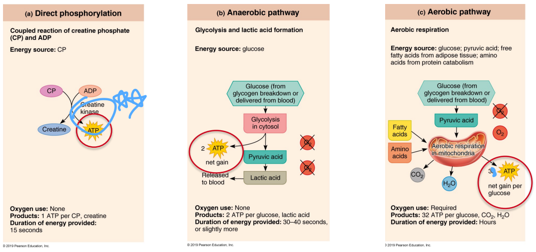

CP + ADP = ATP
(1ATP/CP)
(15s)

ATP + CP = Glycolysis
(2 ATP/glucose =quick)
(30-40s)

Glucose +FFA's + AA's
32 ATP/glucose
(hours)

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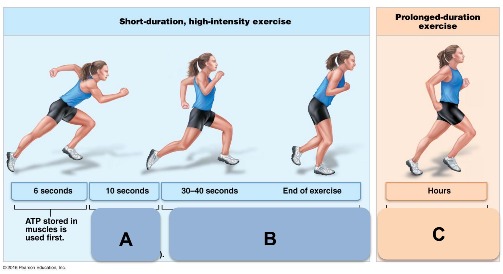
Muscle Metabolism



Activity: Name the 3 sources of ATP production in A, B & C?

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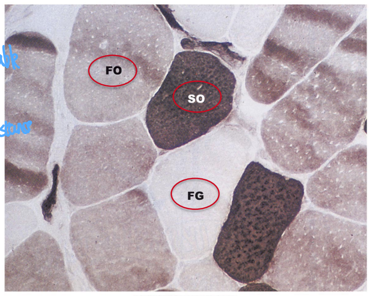
Muscle fibre types

ATP, ↑force, fatigable

1. Fast twitch glycolytic
   - **white**
2. Slow oxidative fatigue resistant marathon
   - **red**
3. Fast oxidative
   - **pink** sprinter



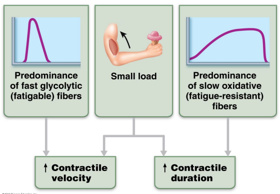
Muscle fibre types

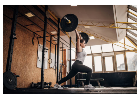
1. **Fast twitch glycolytic**
   - white (fatigable**)**
2. **Slow oxidative fibres**
   - red (fatigue-resistant)

In between fibres?

3. **Fast oxidative** (Pink)

- **Combinations recruited for different actions**





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rel. between contraction speed & ATP synthesis

Muscle Fibres: Characteristics

metabolic features

structural features

Table 9.2 Structural and Functional Characteristics of the Three Types of Skeletal Muscle Fibers

	SLOW OXIDATIVE FIBERS	FAST OXIDATIVE FIBERS	FAST GLYCOLYTIC FIBERS
Metabolic Characteristics			
Speed of contraction	Slow	Fast	Fast
Myosin ATPase activity	Slow	Fast	Fast
Primary pathway for ATP synthesis	Aerobic	Aerobic (some anaerobic glycolysis)	Anaerobic glycolysis
Myoglobin content	High	High	Low
Glycogen stores	Low	Intermediate	High
Recruitment order	First	Second	Third
Rate of fatigue	Slow (fatigue-resistant)	Intermediate (moderately fatigue-resistant)	Fast (fatigable)
Activities Best Suited For			
	Endurance-type activities—e.g., running a marathon; maintaining posture (antigravity muscles)	Sprinting, walking	Short-term intense or powerful movements, e.g., hitting a baseball
Structural Characteristics			
Fiber diameter	Small	Large*	Intermediate
Mitochondria	Many	Many	Few
Capillaries	Many	Many	Few
Color	Red	Red to pink	White (pale)

*In animal studies, fast glycolytic fibers were found to be the largest, but this is not true in humans.

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Activity: Review & Learn

1. Define motor unit
2. 3 types of muscle fibres (fast glycolytic, slow oxidative, and fast oxidative) rely on different cellular processes to obtain ATP.
   What are these processes, what are the main differences between them.
3. FG muscle fibres are engaged in short duration activities whereas SO muscle fibres are engaged in long duration activities.
   What characteristics do these fibres have?

Summary of key concepts: Muscular System 2.5b

- Define **hypertrophy, atrophy and hyperplasia** with respect to skeletal muscle.
- **3 types of muscle fibres** (fast glycolytic, slow oxidative, and fast oxidative) rely on different cellular processes to obtain ATP.
- Cellular processes include use of **direct phosphorylation, anaerobic pathway and the aerobic pathway** to produce ATP.
- **FG** (white) fibres are engaged in short duration activities
- **SO** (red) fibres are engaged in long duration activities